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RESEARCH ARTICLE

Design methodology of digital sensors for detecting laser fault injection attacks in FPGAs

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Abstract

Laser fault injection (LFI) refers to a serious attack that modifies programs and data in embedded systems by introducing specific errors into a device in operation. As a countermeasure, in addition to duplication and optical sensors, the detection of LFIs using a sensor composed of digital circuits has recently been proposed. However, the research on the physical layout of these sensors is limited. In this work, we profile the laser irradiation position and its effect on the surrounding circuits by laser-scanning experiments on field-programmable gate arrays (FPGAs). Studies have shown that digital sensors with conventional designs have low-sensitivity areas with regard to LFIs. To improve LFI detection in FPGAs, we propose a design method for digital sensors and apply it to a ring oscillator (RO)-based digital sensor and a time-to-digital converter-based sensor. Experimental results show that the RO-based sensor based on the proposed method increased the area detecting LFI to about twice that of the conventional method, and detect 99.8% of the effective faults at the target register under protection. Furthermore, we extend the application of the RO-based sensor based on the proposed method to the protection of cryptographic circuits. The results of the laser-scanning experiment show that the proposed sensor detected 98.3% of the LFIs that caused effective faults in the cryptographic circuit.

Keywords Fault injection attack · Laser fault injection · Ring oscillator · Time-to-digital converter · Field-programmable gate array · Digital sensor

1 Introduction

In today's society, electronic devices have various applications across a wide range of fields. The integrated circuits (ICs) embedded in these devices require certified security not only in the logical layer but also in the physical layer because they can fall into the hands of malicious users. Fault injection attacks (FIAs) are physical attacks wherein faults are injected into the running devices to alter the behavior of security functionalities and leak confidential data.

In 1997 when Boneh et al. proposed a method for analyzing secret information by generating faults during the Chinese Remainder Theorem (CRT) algorithm in RSA calculations, FIAs became a realistic threat to cryptographic systems. Correspondingly, various fault analysis methods have been proposed, such as persistent fault analysis [1], ineffective fault analysis [2], and statistical ineffective fault analysis [3].

In the FIA scenario, the attacker injects faults into a device by various means, such as clock glitching, power glitching, electromagnetic fault injection (EMFI), and laser fault injection (LFI). Although clock glitching and power glitching do not require special equipment, their effects are spread over the entire IC; thus, the fault injection location cannot be controlled, and these faults are relatively easy to detect. Conversely, EMFI and LFI feature high selectivity in terms of fault injection location.

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LFI has high spatial and timing resolutions, making it a highly effective method for injecting faults. Subjecting the transistor layer of an IC to laser irradiation causes bit flip in the memory or gate output, which can have serious adverse effects on the system running in the device, including the corruption of internal data, fetching instructions [4, 5], and the circuit configuration on the field-programmable gate array (FPGA) [6]. These effects can result in the leakage of secret keys [7] and the alteration of security processes [8], severely compromising the device security. Therefore, developing effective countermeasures against LFIs has become one of the important goals in the hardware-security research community.

In [9], it is noted that the number of transistors exposed to laser spots has increased with the recent component miniaturization, and the shorting of multiple transistors causes transient IR-drop. This increases signal delay on the chip, causing timing violations. Therefore, countermeasures have been proposed to detect abnormal changes in signal delay due to LFIs. He et al. [10] proposed a ring oscillator (RO)-based sensor that detects LFI by monitoring the oscillation frequency ripple. Ebrahambadi et al. proposed a digital sensor with a time-to-digital converter (TDC) for laser detection [11]. Muttaki et al. proposed a universal sensor that detects disturbances due to FI using the difference between the critical paths of two buffer chains [12]. These sensors monitoring signal delay are composed of digital circuits; as such, they are called digital sensors. Compared to conventional optical sensors, digital sensors have been attracting attention recently because of their low cost, concealment, and portability advantages.

Thus, they are expected to be practical and universal tools for combating FIs, particularly in FPGAs. However, they have some drawbacks in actual applications.

Layout design for LFI detections on FPGA Discussing sensor architecture and layout is essential for protecting security critical circuits from LFI. Several studies have explored methods for designing the layout of digital sensors. [13] presented a layout design for an RO-based sensor in ASIC to protect the Advanced Encryption Standard (AES) from LFI. [14] modified the SoFI framework [15] for FPGAs to identify subcircuits vulnerable to FIAs and designed sensor layouts based on mixed-integer linear programming.

Sensor evaluation based on comprehensive parameters and common metrics.

Existing LFI sensors are assessed uniquely; thus, their effectiveness has not been compared. [10] evaluated the effectiveness of an RO-based sensor through laser-scanning experiments, with randomly selected laser intensity and pulse width parameters. In contrast, [11] verified the effectiveness of a TDC-based sensor through simulations based on a simplified model; however, its effectiveness was not evaluated on an actual device.

We discuss the appropriate physical layout of digital sensors in FPGA. The contributions of this work are as follows:

- To elucidate the relationship between the position of the laser spot and its impact on the signal delay, we investigate the shift in the oscillation frequency of the RO during laser irradiation through laser-scanning experiments.
- We propose a floorplanning methodology for digital sensors that is appropriate for LFI detection on FPGAs, and we applied the proposed method to RO- and TDC-based sensors.
- We assessed the effectiveness of the proposed design by performing detailed laser scans and intensity- and pulse-width-aware laser scans on the RO- and TDC-based sensors. The proposed RO- and TDC-based sensors identified 99.8% and 85.4% of all observed faults in the detailed laser scan, respectively.

The original version of this work has been presented in [16]. This paper contains the following additional content and revised content: Proposal of expanded architecture for the RO-based sensor using a carry chain on FPGAs; (ii) an experiment to evaluate the effectiveness of the proposed RO-based sensor in protecting the AES circuit.

The remainder of the paper is structured as follows: Sect. 2 describes the background knowledge of the study and Sect. 3 reports the results of experiments to analyze the relationship between the laser irradiation position and its effect on the signal delay. Section. 4 proposes an effective digital sensor layout for laser detection and Sect. 5 evaluates the effectiveness of the proposed scheme through laser-scanning experiments. Sect. 6 extends the proposed RO-based sensor and discusses its effectiveness when applied to scenarios aimed at protecting cryptographic circuits. Finally, Sect. 7 concludes this paper.

2 Background

2.1 FPGA

FPGAs have reconfigurable hardware and are applied in various fields because of their superior performance, energy efficiency, fast time-to-market cycle, and design update capabilities. Most recent FPGAs employ static random-access memories (SRAMs), and any desired logic can be implemented by writing the circuit data to the internal SRAM of the chip.

In this work, we utilize Kintex-7, a middle-class SRAM-based FPGA from Xilinx. Figure 1 shows a schematic of the Kintex-7 architecture. The configurable logic blocks

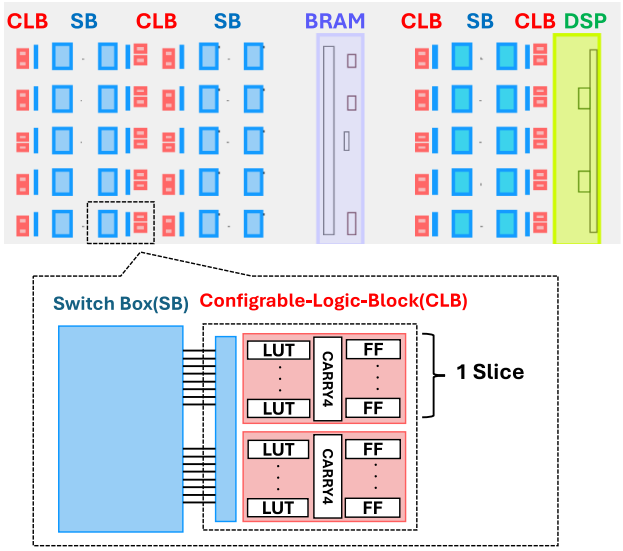


Fig. 1 Overview of the Kintex-7 Architecture

(CLBs) include two slices, each composed of four 6-input/2-output look-up tables (LUTs), eight flip-flops (FFs), and carry chains. CARRY4 is a logic function that calculates carry bits of addition and subtraction at high speed, consisting of four MUXes and four XOR gates. Switch boxes (SBs) manage the interconnections between each CLB and other functional elements. In addition to these standard component, the FPGA can utilize dedicated circuit resources such as block RAMs (BRAMs) and digital signal processors (DSPs).

2.2 Laser fault injection

Optical fault injection in a security context was explored by Skorobogatov et al. in 2002 [17]. As outlined in their work, when laser light impinges on the p–n junction, the photoelectric effect engenders an electron–hole pair, triggering a transient short circuit within the transistor. For devices with relatively small process sizes, an additional current component is generated, causing a broad drop in the supply voltage [9]. LFI can induce bit flips in SRAMs [18], setup time violations caused by voltage drops [9], and read errors in flash memory [5].

Research has been conducted on laser faults in FPGAs. As discussed previously, the SRAM cells within FPGAs store values that govern the logic and routing of the circuit. LFIs can flip these SRAM values, thereby altering the circuit configuration. This fault persists until the FPGA is reprogrammed with the circuit, severely impacting the operation of the system implemented on the FPGA. Recently, an analysis method known as [1] was proposed, which estimates keys using enduring faults in S-box tables. This renders both temporary and persistent faults as substantial threats.

2.3 Digital sensors

Recently, many countermeasures against FIA have been proposed using sensors composed entirely of digital circuits. These sensors use delay lines (buffer chains or ROs) to convert the chip power-supply voltage into a signal delay and measure its fluctuations. Therefore, disturbances in the power-supply voltage caused by FIAs, such as LFI, EMFI, and power glitching, are captured as abnormal changes in the signal delay of the delay line. These sensors are easier to design and calibrate and offer better concealment capability and portability than specific analog sensors, such as photodetectors and bulk current sensors [19]. Additionally, unlike classical duplication-based countermeasures [20], these sensors detect LFI independent of the internal state of the target, thus protecting cryptographic circuits from attacks such as IFA and SIFA that do not require faulty ciphertext. In this section, we describe two types of sensors for LFI detection.

2.3.1 RO-based LFI sensor

The RO-based method monitors the state of the RO and detects abnormal changes in the supply voltage due to intentional disturbance injection. There are two types of RO-based schemes: the first counts the oscillation frequency [21], whereas the other monitors the phase change of the frequency [10]. The former requires a wide measurement interval to maintain sufficient accuracy and is not suitable for the rapid detection of LFI. This study mainly focuses on the latter.

Figure 2 shows the architecture of the RO-based sensor proposed by Wei et al. The sensor consists of an RO with a minimum of three inverters, a disturbance capture with two FFs to monitor its state, and a delay factor to insert a delay of one cycle. Three signals ($f1$, $f2$, and ck) are extracted from each of the three inverters of the RO, and $f1$ and $f2$ are sampled by ck_delay.

As shown in Fig. 3(a), under normal conditions, the falling edge of ck_delay is located between the rising edges of $f1$ and $f2$; thus, $Q1 = 1$, $Q2 = 0$, and $Alarm = 1$ (non-detection). Conversely, as shown in Fig. 3(b-c), if the oscillation frequency is abnormal due to an external disturbance, the values of $Q1$ or $Q2$ are flipped and $Alarm = 0$ (detection) are set. The sensor's sensitivity is determined by the phase difference between $f1$ and $f2$ (time window for sampling: TWS). The narrower the TWS, the smaller the shift in the oscillation frequency that triggers an alarm.

2.3.2 TDC-based LFI sensor

The TDC functions as a sensor that samples the signal delay of a delay line, transforming it into a digital value. Figure 4 depicts a schematic of the TDC. The figures present the initial delay line (IDL) composed of cascaded n_0 buffers and

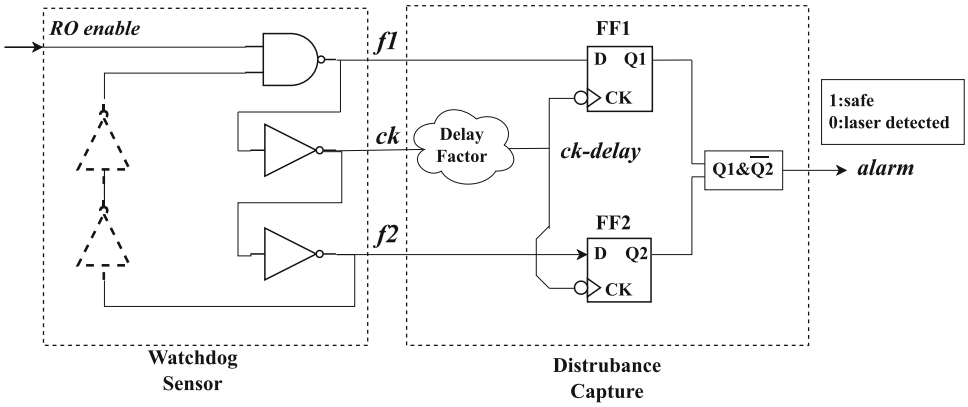


Fig. 2 RO-based laser sensor

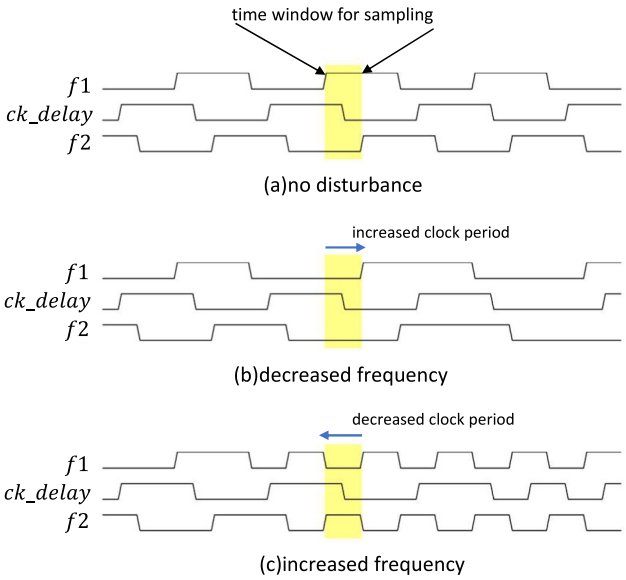


Fig. 3 Timing-chart in disturbance capture

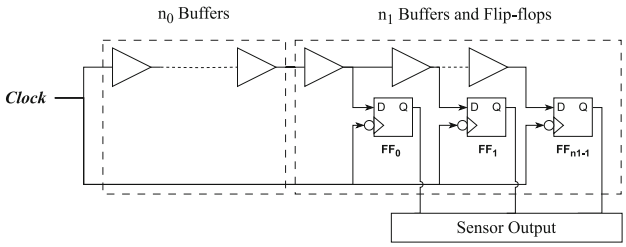


Fig. 4 The architecture of TDC

a tapped delay line (TDL) composed of n_1 buffers and FFs. The IDL is used to delay the input signal by one clock cycle. The TDL is used to assess the delay at the FF position, where an incoming clock signal incites a setup time violation. The shorter the signal delay between buffers in the TDL, the better the resolution of the signal delay. $n1$ corresponds to the number of output bits from the TDC. The resolution, along with $n1$, determines the quantum of the signal delay at which output saturation occurs.

The output of the FFs is inputted to some post-process, such as comparison with a threshold. In reference [11], to suppress false positives, the post-process is configured to trigger an alarm when the difference between the output of the current cycle and the average of some previous cycles become so large that it exceeds the threshold. [22] compared the power waveform of the golden model with the TDC output and used a machine learning model to identify abnormal fluctuations. In this study, we used a constant value as the reference to detect LFI, given the low on-chip voltage fluctuation and the steady TDC output.

3 Profiling the effects of laser injection

To implement effective digital sensors for LFI detection, it is crucial to clarify the relationship between the location of laser injection and the resulting changes in signal delay near the laser spot. Here, we investigate the effects of laser irradiation on the IR-drop and signal delay variation using laser-scanning experiments on Kintex-7 FPGA, a 28-nm-process FPGA, and discuss the physical design of an appropriate sensor.

3.1 Experimental setup

Figure 5 shows our experimental setup. To irradiate a laser from the backside of the chip, a PDM+ laser source with a wavelength of 1064 nm, provided by Alphanov, was used. This device can emit a maximum power of 1.35 W at a voltage of 5 V. A 20x objective lens is selected, and a pulsed laser with a spot diameter of 4 µm is emitted through the optical system. The timing of injection and the pulse width were set in 10 ns steps by the control FPGA (Artix-7). We chose the SAKURA-X (SASEBO-GIII) board [23] as the target device. This board is equipped with a flip-chip-bonded 28-nm FPGA, Kintex-7, as the evaluation target. The board is

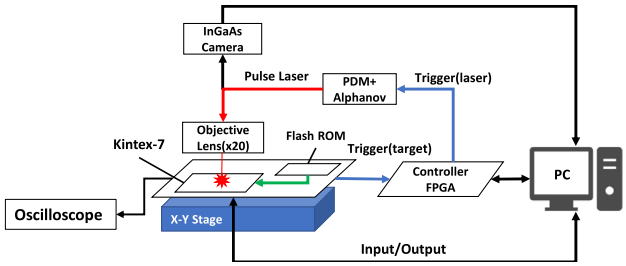


Fig. 5 Schematic of the experimental setup

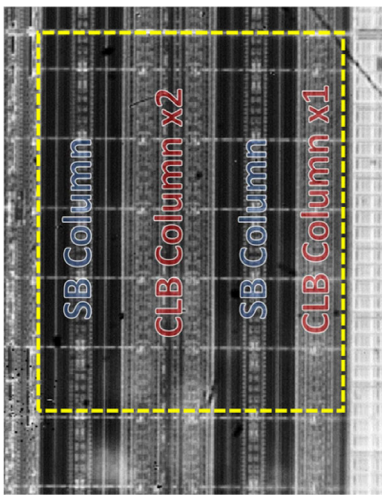


Fig. 6 Image of the area where the evaluation target was mounted

fixed to the XY stage, and the laser irradiation position is adjustable in 4 μm increments. Further, an indium gallium arsenide camera was adopted to observe the structure of the target FPGA where the laser was injected.

Figure 6 shows an image of a part of the Kintex-7 FPGA observed with a 20x objective lens. Five columnar patterns can be observed: SB ×2, CLB ×2, SB ×2, and CLB ×1 from left to right. Since LFI may modify the target's circuit configuration, the same input was applied again after each injection to verify that there were no persistent faults affecting the output. Thereafter, the circuit was reconfigured to remove the errors caused by the LFI.

3.2 Target Circuit

This section describes the circuit used in the experiments. The experiments use a RO-based sensor and a target circuit to be protected. Figure 7 shows the layout of the circuit under evaluation, implemented in Xilinx's FPGA development environment, Vivado 2020.2. The RO sensor, consisting of three inverters (highlighted in blue) and an FF (highlighted in purple), is implemented around the 128-bit register circuit (highlighted in red) to be protected. The signal f1 and the alarm signal are outputted from the GPIO pins on the board,

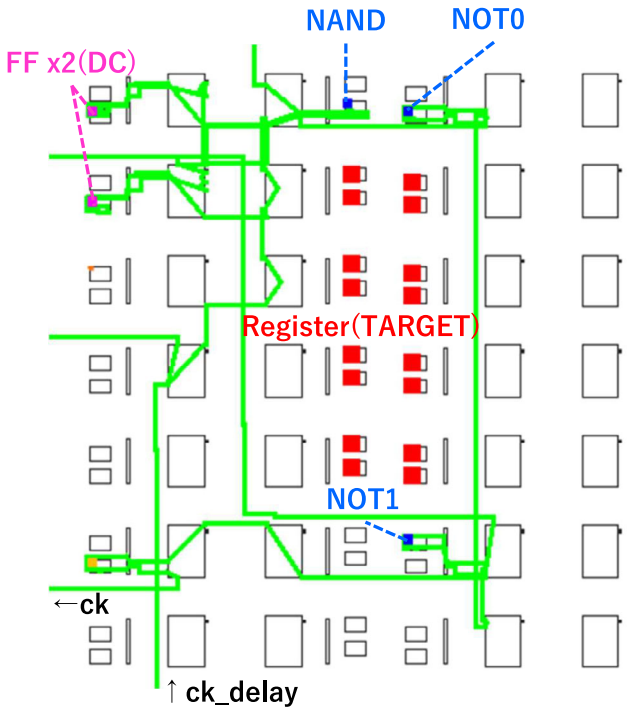


Fig. 7 Layout of a simple RO-based sensor

and the oscillation frequency is recorded by an oscilloscope. When the target circuit was implemented in the experimental environment, the RO oscillated at 482 MHz.

3.3 Laser-scanning experiment

Figure 8 presents the oscillating frequency of the RO at each point. This section describes the laser-scanning experiment on the target circuit.

The laser-scanning experiment was performed on 100 × 120 points in the yellow area in the figure. The laser intensity was 1.35 W, and the pulse width was 300 ns. The oscillation frequency of the RO was recorded at each irradiation position with a width of 200 ns at 100 ns after the rise of the laser pulse.

Figure 8 shows a mapping of the measured oscillation frequency of the RO at each laser irradiation position. From these results, the effects of LFI on the oscillation frequency can be classified into three categories: (1) decrease and stop, (2) increase, and (3) no effect.

(1) Decrease in frequency: It has been observed that the oscillation frequency decreases to approximately 400 MHz, predominantly around the SB columns. This result is consistent with the experimental results on Virtex-5, as presented in [9]. Therefore, this frequency drop may be attributed to the IR-drop by LFI. In some instances, the oscillation ceases at the CLB where the inverter is situated, as well as at the SB columns. This phenomenon may be attributed to the modi-

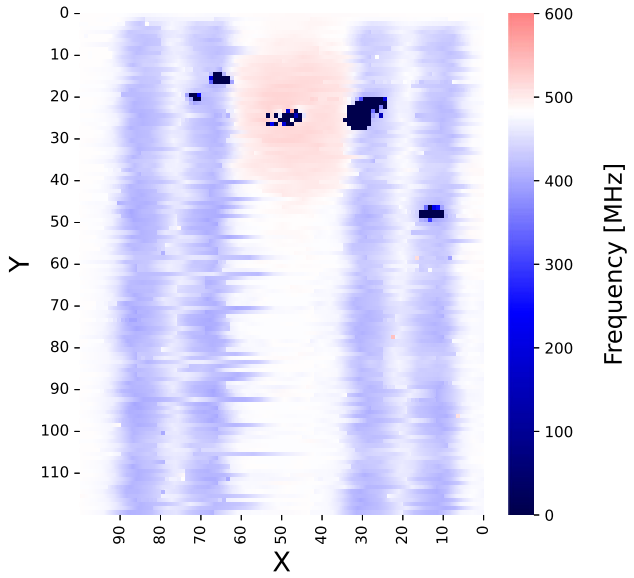


Fig. 8 Oscillating frequency of the RO during laser irradiation at each point

fication of the RO's configuration or the temporary stacking of the inverter output.

(2) Increase in frequency: An area of increased oscillation frequency was observed around the inverters placed adjacent to each other. This rate of increase decreases as one moves away from the inverter, suggesting that this is a local phenomenon occurring only around the LUT irradiated by the laser. Interestingly, the frequency fluctuations occurred even when the LUT was not directly hit. Xilinx has not revealed the details of the device; thus, the mechanism of this phenomenon remains unknown. However, we can utilize the results for designing digital sensors in FPGAs.

(3) The findings highlight a potential limitation of conventional digital sensors in accurately detecting the laser irradiation of CLBs. Laser irradiation of CLBs can only be detected when the oscillation ceases or the frequency increases, both of which occur only around the inverter of the RO. Consequently, if the LUTs comprising the sensor are positioned uniformly, it may be relatively straightforward for an adversary to circumvent the sensor and inject a fault into the protected target on the CLB. Nevertheless, the floorplanning of the sensor and protection of the target circuit on FPGAs have received limited research attention. For instance, in the illustrative implementation of [10], the RO comprises merely three inverters, which are insufficient to encompass the CLB where the target is placed.

4 Proposed method

As discussed in Sect. 3.3, if the sensor and the target circuit to be protected are placed far apart, an attacker can inject faults into the target while avoiding the sensor. In this section, we propose a method for arranging the components of a sensor that can protect not only the SB columns but also the CLBs, and we explore its application in two types of digital sensors.

4.1 Design methodology

TDC- and RO-based sensors comprise a delay line (RO or buffer chain) and a sampling circuit designed to monitor the signal delay from the delay line's output. The physical layout of the former pertains to the area the sensor covers, whereas the latter corresponds to the sensor's sensitivity. In this section, we introduce a design methodology for the physical layout of these two circuits.

Initially, we present a design methodology for the digital sensor's delay chain. The most crucial aspect when designing the logic chain involves the positional relationship between the logic elements of the target and those of the sensor. As shown in sect. 2.1, Xilinx FPGAs have multiple LUTs per slice. We propose to reserve one of them for the sensor delay lines and the rest for the security critical user logic. Our placement policy forces the target user logic and sensor delay lines to be physically close together. As a result, the effect of LFI on the target is more likely to be mirrored in the output of the delay line. For instance, If the delay line's logic elements are exposed to laser radiation directly, its output could be transiently or persistently faulted, leading to an abnormal signal input into the monitoring circuit. Even if the laser does not directly irradiate the LUTs of the delay line, it will cause a change in the signal delay of the delay line as shown in sect. 3.3, and the monitoring circuit will detect the abnormal change.

Next, we describe the design policy for the circuit that monitors the signal delay at the output of the delay line. The physical layout of the monitoring circuit determines the jitter threshold that determines whether an alarm is triggered. The smaller the jitter threshold, the more sensitive the alarm output is to environmental changes. Therefore, our design policy is to adjust the physical layout of each logic element and the signal delay between them so that the jitter threshold of the sensor is sufficiently small to detect LFI.

4.2 RO-based sensor with inverter array

Figure 9 illustrates a diagram of the RO-based sensor developed using the proposed methodology. Proposed RO-based sensor has an ring oscillator with additional inverter array consisting of N inverters to protect N CLBs against LFI. Since the number of inverters of the ring oscillator must be odd, if N +3 is even, a inverter must be replaced by a buffer. To preserve the phase difference between $f1$ and $f2$ ie., TWS, the inverter array is inserted between a NOT gate and a NAND gate that supply RO signal to the DC. These inverters are regularly placed to form an array that is sensitive to

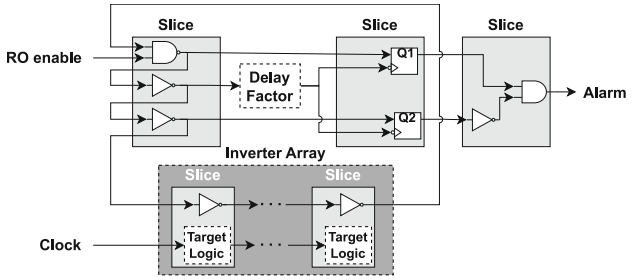


Fig. 9 Architecture of the RO-based sensor based on the proposed method

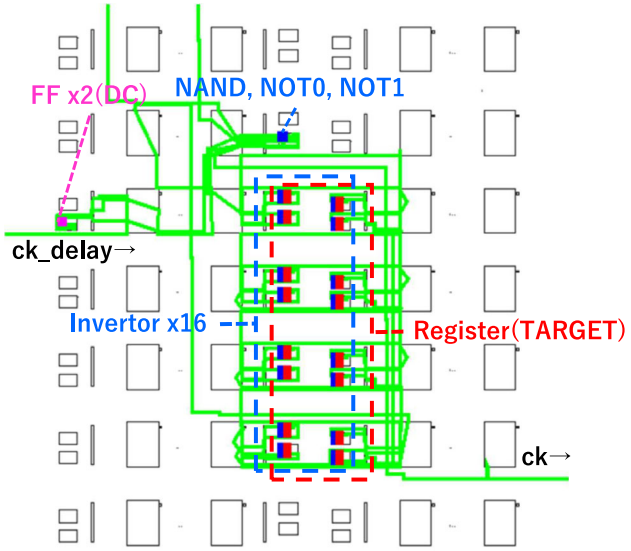


Fig. 10 Example of the RO-based sensor

LFI. All or a portion of the security critical LUTs and FF of the user design is situated within this array. Since TWS is determined by the signal delay between a NAND gate and the NOT gate, these inverters are placed in a single slice, and the two FFs of DC are located in one slice to minimize the TWS.

There is a trade-off between the number of slices that the sensor protects and its sampling rate. The more inverters in the ring oscillator, the more the number of slices that can be protected, but the lower the oscillation frequency. Since two FFs in disturbance capture is driven by the ring oscillator signal, the oscillation frequency corresponds to the sampling rate of the sensor. If you need to protect more slices and to keep the sampling rate, another RO-based sensor must be added.

Figure 10 demonstrates an example of an enhanced RO-based sensor with a 128-bit register as a target circuit of LFI. Nineteen inverters form the RO: three inverters (NAND, NOT0, NOT1) which supply the RO signals to DC are placed at one slice, sixteen inverters are placed in each of the 16 slices in which the register is implemented. The arrangement of the three inverters minimise the phase difference between $f1$ and $f2$. This arrangement enable the RO-based sensor to detect frequency ripples equivalent to the delay time through a single LUT. The delay factor consists of four inverters, with a delay equivalent to several cycles of the oscillation frequency inserted by modifying the placement and wiring of the inverters. When this circuit was implemented in the device, the oscillation frequency dropped to 80 MHz, due to the increased signal delay necessary to propagate through the entire RO. According to Vivado power simulations, the dynamic power consumption of the target with RO-based sensor (sum of signals clocks and logic) is lower than 4mW.

4.3 TDC-based sensor with a buffer array

Figure 11 illustrates a diagram of the TDC-based sensor developed using the proposed methodology. The initial delay line of the TDC consists of several buffers that are interconnected in series. In our proposed method, these buffers are arranged one per slice in a regular pattern of vertical, horizontal, or matrix. The security-critical LUTs and FFs are placed on the buffer array and protected from laser injection. The sampling circuit is designed such that the signal delay between buffers of the tapped delay line is sufficiently small to detect the clock jitter that increases owing to laser irradiation. Note that the number of slices that are covered by the TDC is limited because the length of the delay line (number of buffers) needs to be adjusted to match the clock period. Therefore, if you need to increase the number of slices to be protected, it is need to add another TDC or to drop the clock frequency.

Figure 12 presents an example implementation of the TDC sensor to protect a 128-bit register. Although TDC can be implemented using carry chains, the output becomes unstable during non-attacks if the resolution is excessively high. Consequently, this study utilizes a LUT that outputs the input value instead of a carry chain. The TDC is supplied with a 24-MHz clock signal, and the TDC delay line is implemented as 95 buffers. Each slice accommodates one buffer, forming a buffer array. The sampling buffers and FFs are located in a single CLB, and the output capture a signal delay variation equivalent to the delay of one LUT. The TDC output is inputted to a comparator, located outside Figure 12, and compared with the previously recorded TDC output. An alarm is triggered if the output and reference values do not match. Upon implementation of this circuit, '00000111' was outputted from the TDC stably during the steady state. According to Vivado power simulations, the dynamic power consumption of the target with TDC-based sensor (sum of signals clocks and logic) is lower than 3mW.

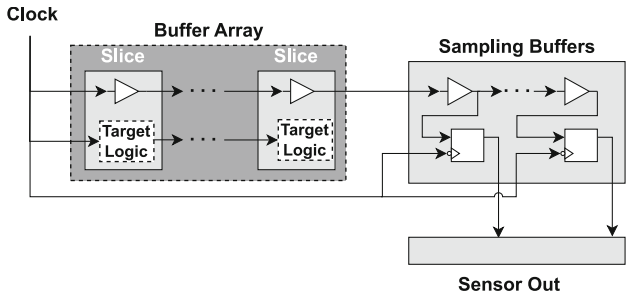


Fig. 11 The architecture of the TDC-based sensor with the proposed method applied

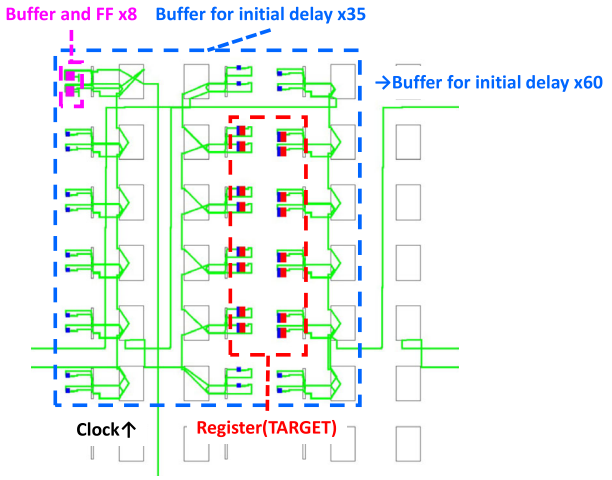


Fig. 12 Example of a TDC-based sensor

4.4 Overhead

This section elucidates the overhead associated with the proposed physical layout design for digital sensors meant for LFI detection.

Initially, we consider the logic resources consumed by the RO-based and TDC-based sensors based on the proposed methodology. Table 1 illustrates the logic resources required to protect N slices with M sensors. Here, $n1$ denotes the number of output bits of the TDC sensor, and C is the logic resource consumed by the comparator of a single TDC sensor. For both types of sensors, the logic resources consumed increase linearly with the number of slices to be protected and the number of sensors. Subsequently, we discuss the increase in the number of slices (area penalty) resulting from the application of the proposed method. In Kintex-7, four 6-input-2-output LUTs are available per slice. Within a sensor array, one of the four LUTs is consumed for the sensor, thus reducing the available LUT resources per slice to 3/4. Consequently, the area penalty of implementing a circuit on a sensor designed based on the proposed method is 33%.

Table 1 Logic resources consumed by the M sensors to protect N slices

	LUT	FF
RO-based	N + 5M	2M
TDC-based	N + n_1*M + C*M	n1*M

5 Evaluation

In this section, we verify the effectiveness of our method through two types of laser-scanning experiments. First, we describe a detailed laser-scanning experiment that was conducted with fixed intensity and pulse width. Second, we report the results of a further laser-scanning experiment that was conducted to investigate the detectable laser parameter sets for the proposed RO- and TDC-based sensors.

5.1 Detailed laser-scanning experiment

In this section, we evaluate the effectiveness of the proposed methodology through detailed laser-scanning experiments. LFI was implemented on 100 × 120 points in the area within the yellow frame (400 µm x 480 µm) in Fig. 6, with a pulse width of 300 ns, and the spot positions (x,y), register outputs, and alarm outputs were recorded.

Based on the obtained outputs, the coordinates were classified into the following four classes.

- $alarm$: The alarm is triggered, and the output of the register is not faulted.
- $fault\&alarm$: The alarm is triggered, and the output is faulted.
- $fault$: The alarm is not triggered, and the output is faulted.
- no_effect: The alarm is not triggered, and the output is not faulted.

Next, we discuss two indicators of sensor detection performance: alarm rate and detection rate.

The alarm rate is the percentage of the area that is successfully detected in the scanning area, and the detection rate is the percentage of the LFI coordinates that are successfully detected by the sensor with faults in the register output.

$$alarm_rate = \frac{|alarm| + |fault\&alarm|}{ALL} \quad (1)$$

$$detection_rate = \frac{|fault\&alarm|}{|fault\&alarm| + |fault|} \quad (2)$$

where ALL is all the coordinates that have been scanned.

The table summarizes the results of the laser-scanning experiment. The improved RO-based sensor improves the $alarm_rate$ to approximately twice that of the conventional

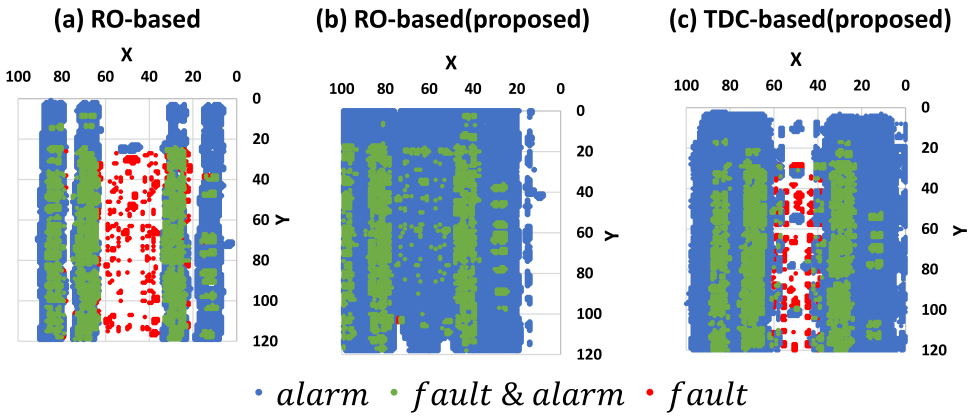


Fig. 13 Fault mapping of the two sensors based on the proposed layout and a sensor based on the conventional layout

one and detects 99.8% of valid faults. Conversely, the TDC-based sensor achieved an improved *alarm_rate* compared to that of the conventional sensor; however, its *detection_rate* did not improve as considerably as that of the RO-based sensor.

Figure 13 shows the three maps obtained in the experiment. Figure 13(a) is consistent with the results of the frequency variation map presented in Sect. 3.3. These results show that the RO-based sensor with a naive layout is sensitive to LFI in the SB columns, whereas the LFI into the CLB matrix (X = 0–5, 35–62) does not disturb the oscillation frequency and does not trigger the alarm even though an effective fault is injected into the register.

Figure 13(b) shows that the RO-based sensor based on our method triggers alarms in both the SB column (X = 20–49, 50–77) and the CLB column (X = 0–19, 50–77), indicating that it can cover almost the entire target. In particular, the CLB column successfully detected lasers in 95.8% of the area and 98.2% of valid faults.

Figure 13(c) shows that the TDC-based sensor based on our method increase the number of coordinates in the SB columns where alarms are triggered (X = 6–34, 63–90). Conversely, when limited to the CLB column (X = 35–62) with the register, the percentage of coordinates where alarms were triggered was 34.6%, and 24.6% of the effective faults were detected. We believe that this result was obtained because the TDC consisting only of LUTs did not have a sufficient resolution, and we can improve the *alarm_rate* and *detection_rate* in CLB columns by adopting a higher-sensitivity TDC with a carry chain.

5.2 Further laser-scanning experiment

In this section, we assess the detection performance of the sensors based on our method, considering the laser intensity and pulse width in addition to the irradiation position. The laser scanning is conducted at a total of 11 × 13 points. The voltage supplied to the laser source varied in five steps, from 1 V to 5 V, and the pulse width was adjusted in five steps from 20 ns to 100 ns at each irradiation position. To accommodate the effects of environmental noise and input values, the laser is emitted 50 times with the input values randomly altered each time. Additionally, to evaluate the response time of the sensors, the delay between the trigger and the alarm was logged.

First we evaluate the fault sensitivity of the register to be protected at each laser parameter. Figure 14 shows the number of register faults in intensity-pulse width for the laser scanning experiment. The results show that the higher the intensity and pulse width, the more faults are injected into the register. No faults were injected when the intensity was less than 2 V or the pulse width was less than 20 ns.

Next, we evaluate the laser detection performance of the RO and TDC sensors. Table 3 presents the results of scanning the RO- and TDC-based sensors. Among the 178,750 laser irradiation processes, the RO- and TDC-based sensors detected the laser 68,196 and 50,877 times, respectively. The average delay of the alarm output of the RO-based sensor was 80.3 ns, with the delay tending to be smaller in the inverter array than in the SB column. Conversely, the average delay of the TDC sensor was 97.6 ns, signifying that its response time is longer than that of the RO-based sensor.

Figure 15 illustrates the number of detections by the RO- and TDC-based sensors for each intensity and pulse width. The maximum number of detections for each parameter is 7,150. Both sensors display a decline in the number of detections as the laser intensity and pulse width decrease, and the RO-based sensor registers a higher number of detections than the TDC sensor under most conditions. The disparity between the two sensors is particularly prominent at low intensities. Even at an intensity of 1 V, the RO-based sensor detected the laser up to 1,298 times, whereas the TDC sensor failed to detect the laser at any pulse width.

Subsequently, we concentrate on the number of detections for each irradiation position. Figure 16 summarizes the number of laser detections for the two sensors at each coordinate. The maximum number of detections for each coordinate is 1250. X = 2–4, 8–10 are SB columns, X = 0–2, 5–7 are CLB

Table 2 Result of laser scans against the naive layout RO-based sensor and the two sensors based on the proposed layout

	Naive layout RO-based sensor	Improved layout RO-based sensor	TDC-based sensor
Alarm	2,959	7,758	6,935
Fault	378	3	360
Alarm & fault	1,941	1,857	2,110
No Effect	6,722	2,382	3,135
Alarm_rate	40.8%	80.1%	75.3%
Detection_rate	83.7%	99.8%	85.4%

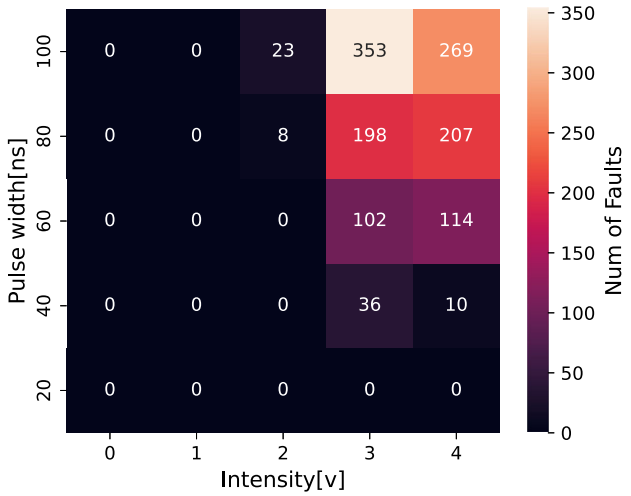


Fig. 14 The number of data faults at each laser parameter

Table 3 Number of LFI detections by the RO- and TDC-based sensor in the further laser-scanning experiment

	RO-based	TDC-based
Detected	68,196	50,877
Not Detected	110,554	12,773
Total	178,750	178,750

columns, and the slices to be protected are located at X = 5–7. Figure 16 (a) indicates that the RO-based sensor has the highest number of detections in the region where the inverter array is situated (Y = 3–7). In contrast, Figure 16 (b) reveals that the TDC-based sensor detects approximately the same number of laser shots in the SB column as the RO-based sensor; however, it identifies few laser shots in the CLB columns.

6 Protecting AES

In this section, we describe an extension of the RO-based sensors for the protection of cryptographic circuits and evaluate its effectiveness through a laser-scanning experiment.

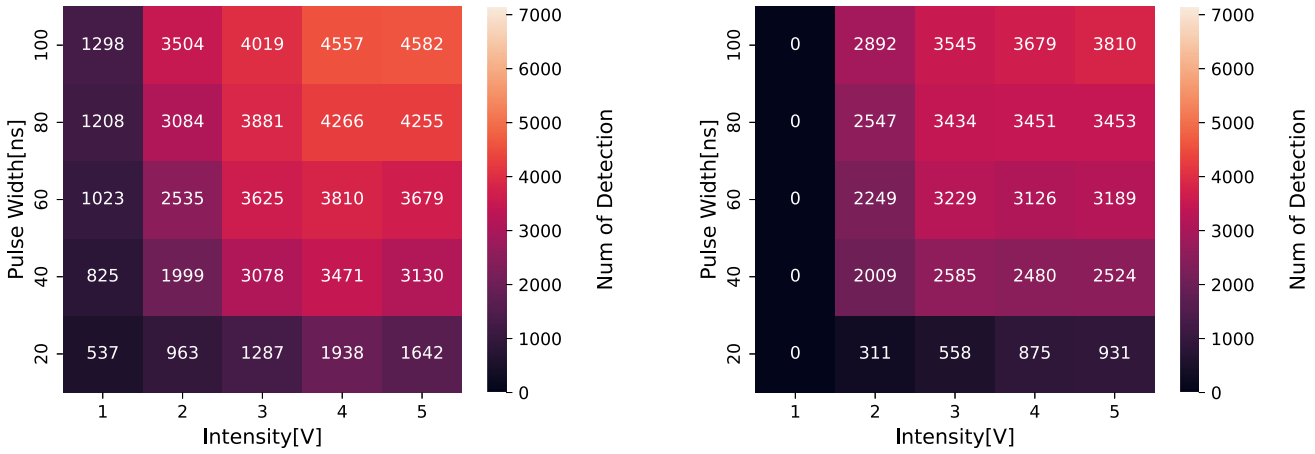
6.1 Improving the sensitivity of RO-based sensors

TDC-based schemes can improve the resolution of signal delays using a carry chain and can adjust sensitivity by modifying the postprocessor and threshold values. However, when an external clock is used, an attacker can disable the sensor by stopping the clock, during which time the circuit configuration can be modified by LFI. Conversely, RO-based schemes can be independently enabled when power is supplied and used as a countermeasure against side-channel attacks. However, the sensitivity calibration of RO-based sensors is difficult because of the modification of the length of wiring between inverters to adjust the width of the TWS. In addition, RO-based methods only exist in variations that use LUTs and have lower signal delay resolutions than TDC-based sensors with a carry chain.

We propose a new architecture that overcomes the limitations of RO-based sensors. The proposed architecture is shown in Figure 17. The sensor consists of a RO embedded with a carry chain composed of M CARRY4s, 4M FFs for sampling the tapped signal, a delay factor, and a postprocessor. This study assumes the use of CARRY4 available in Kintex-7 for the carry chain; however, it can be applied to many modern FPGAs with similar CARRY functions.

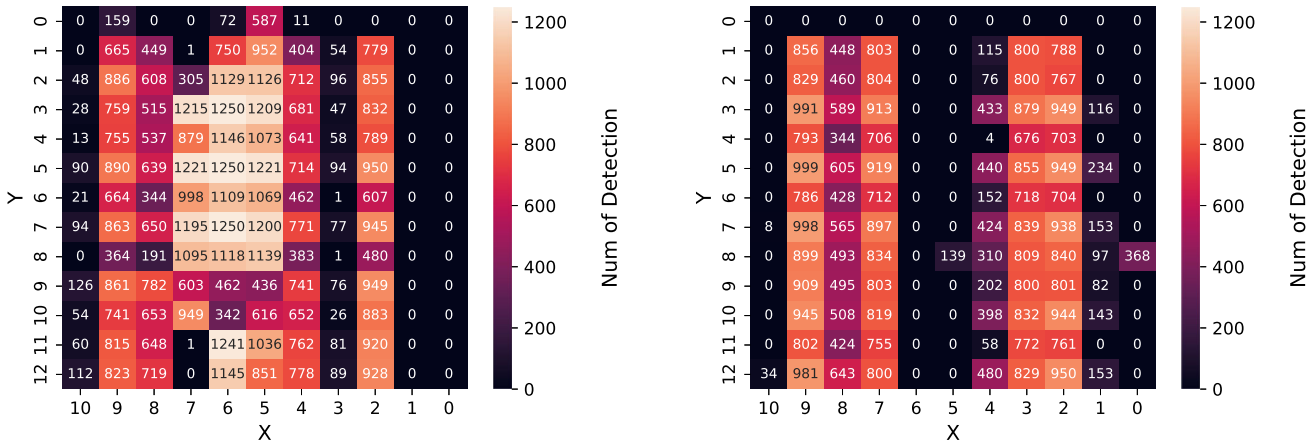
The signals, $\{f_1, ..., f_{4M}\}$, tapped at the carry chain are delayed, corresponding to the number of passing MUXes and the XOR gate, and they are sampled by the FFs driven by ck_delay. The earlier the rising edge of f_0 than that of ck_delay, the more FFs capture "1" because the edge propagates to the later FFs before sampling. Therefore, the Hamming weight of the value sampled by the FFs becomes a digital value representing the phase difference between f_0 and ck_delay. The length of the delay line is adjusted so that half of the FF sample "0" and the other half sample "1" under normal conditions to capture both increasing and decreasing signal delays. The resolution of the signal delay corresponds to one or two MUXes in a carry chain, which is an increase in resolution over conventional LUT-only implementations.

In post-processing, the system detects abnormal environmental changes based on the magnitude of phase change in the RO output from the FFs and triggers an alarm. In the con-



(a) Proposed RO-based sensor (b) Proposed TDC-based sensor

Fig. 15 Number of detections at each pulse width and intensity



(a) RO-based sensor (b) TDC-based sensor

Fig. 16 Number of laser detections at each position

ventional method, it is necessary to adjust the width of the TWS by layout wiring to calibrate the alarm trigger sensitivity. Conversely, using the proposed method, we can acquire digital values of signal delay variations, allowing for the flexible adjustment of alarm trigger thresholds.

6.2 post-process

Since digital sensors measure a signal delay, the outputs depend on the dynamic power consumption due to the logic activity of the target circuit and peripheral. In particular, in practical scenarios where large and complex circuits, such as AES, are protected, the severity of the problem increases. Indeed, it has been demonstrated that digital sensors can be exploited for side-channel attacks in multi-tenant FPGAs [24]. Therefore, when applying digital sensors to actual scenarios, post-processing must be designed to ensure that dynamic power consumption variations due to normal logic operation are not falsely detected as IR drops due to LFI.

In previous studies, methods utilizing the difference between the average value of several cycles and the value of the current cycle or a machine learning model to capture anomalous fluctuations have been explored. In this study, we set a fixed threshold $((T_{sup}, T_{inf}))$ for LFI detection. The $alarm$ is defined as follows.

$$alarm = \begin{cases} 1 & \text{if } T_{sup} < HW(O) | T_{inf} > HW(O) \\ 0 & \text{else} \end{cases} \quad (3)$$

where $HW(O)$ is a hamming weight of the output (O) from the sensor. The threshold values are determined based on the histogram of $HW(O)$ observed when the AES is oper-

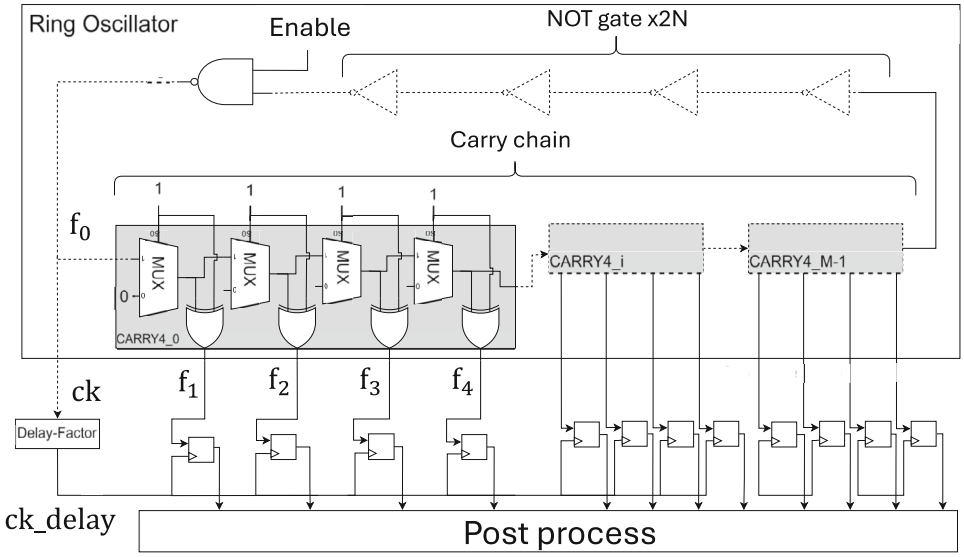


Fig. 17 The architecture of the proposed RO-based sensor

ated multiple times and defines the operating voltage range of the circuit to be protected under non-attack.

6.3 Evaluation target

This section describes the evaluation target used in the laser-scanning experiment. Figure 18 shows the target's layout. It consists of an AES that performs one round of AES calculations per cycle, peripheral input and output circuits, and 16 proposed RO-based sensors. Each instance of the sensor system consists of 60 inverters, 1 NAND gate, 8 CARRY4s, and 32 FFs, and it protects 60 CLBs. Thus, the sensor system protects 480 CLBs in total.

The delay factors are programmable delay chains consisting of selectors, LUTs, and carry chains. The length of each chain is manually adjusted until the appropriate output is obtained from the instance. Although this task requires some effort, it can be automated by adding some additional logic resources.

The postprocessor receives 32-bit outputs (O) from each sensor instance, calculates their Hamming weights ($HW(O)$), and outputs them externally via GPIO pins. Due to the constraints of the experimental environment, we cannot record the outputs of all instances simultaneously. Therefore, one instance is specified for each AES run, and the output values are recorded using an oscilloscope and compared with the thresholds on the host PC. In actual operation, the output values are compared with the thresholds, and if the output exceeds the thresholds, the alarm is triggered.

As discussed in sect. 6.2, the outputs of the sensors fluctuate depending on the operation of the protected object and the peripheral circuits. To determine the sensor thresholds to minimize the risk of false positives, the evaluation target was implemented in an experimental environment, and the outputs of 16 sensors during the AES operation were recorded with 1000 waveforms. Figure 19 shows a histogram of the outputs obtained from each sensor instance during the period from the start of the AES execution to 1 µs. Due to the different inverter configurations, the output trend varied from instance to instance; however, most cases fell in the range of 12–20. The maximum and minimum values obtained from each sensor in the 1000 waveforms are used as threshold criteria for each instance to evaluate the laser detection performance in the following section.

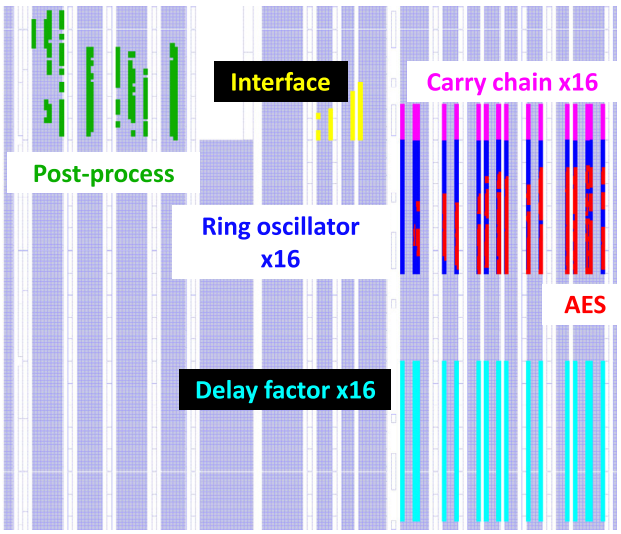


Fig. 18 The layout of the target AES with 16 proposed RO-based sensors

6.4 Experimental result

To evaluate the detection performance of the sensors, laser scanning was performed on 80×85 points within the frame of

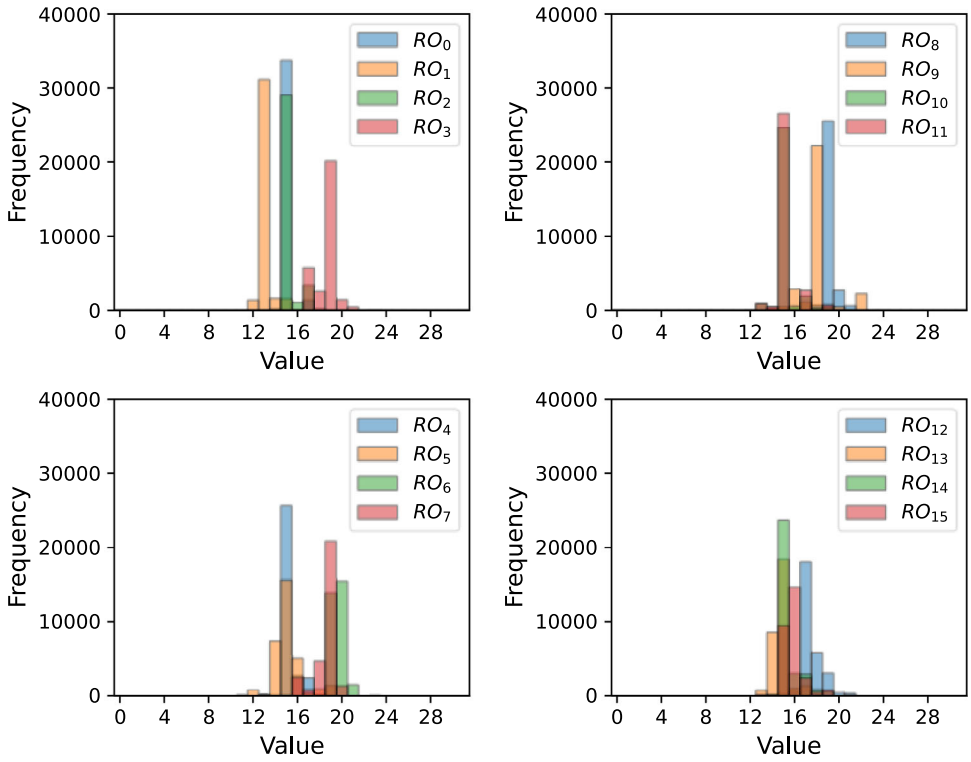


Fig. 19 The histograms for the $HW(O)$ of each RO-based sensor

the sensor where the AES was placed. The laser pulse width was set to 300 ns and the intensity to 1.3 W. The output of each sensor was recorded during a period of 1 µs from the trigger of the circuit under protection. Since only the outputs of one instance per trial can be recorded, 16 LFI trials were performed to record all sensor outputs for each position.

Figure 20 shows a heat map of the minimum and maximum $HW(O)$ outputs from RO_{10} at each irradiation position. We can see a vertical stripe pattern where the output of the sensor changes significantly. The patterns are consistent with the position of the SB columns. In contrast, except for the SB columns, the sensor output changes significantly only in the region of X = 57–60, i.e., the CLBs where RO_{10} is located. Similarly for the other sensors, in the CLBs with the inverters of the ROs, the minimum value of $HW(O)$ tended to be considerably smaller than under non-attacks.

Next, we evaluated the LFI detection performance of the proposed method with the classification and evaluation metrics introduced in Sect. 5.1 The threshold values (T_{sup}, T_{inf}) of each instance were set as the maximum and minimum values that are observed when the AES is running under non-attack in the prior analysis, respectively. A single attack is defined as 16 laser irradiations, and an injection of a fault is identified if at least one error cipher text is obtained among 16 attempts. Furthermore, the LFI is considered to have been detected if one or more sensors trigger an alarm.

Table 4 illustrates the classification result. The $alarm_rate$ is 73.7% and the $detection_rate$ is 98.3%. The result indicates that the proposed sensors are effective for protecting the cryptographic circuit from LFI. Figure 21 depicts the classification for each position. The results demonstrate that the proposed sensor can safeguard both the SB area and the CLBs where the AES is mounted from LFI.

The lower alarm rate in comparison to the scenario presented in Sect. 5.1 may be attributed to the observation that the laser irradiation into the unused BRAM and DSP had a negligible impact on the surrounding signal delay and did not result in a change in the sensor output. BRAMs are equipped with an error correction function, which renders them tolerant to bit inversion due to LFI. However, DSPs that have arithmetic circuits and registers may be susceptible to LFI, necessitating the implementation of additional countermeasures.

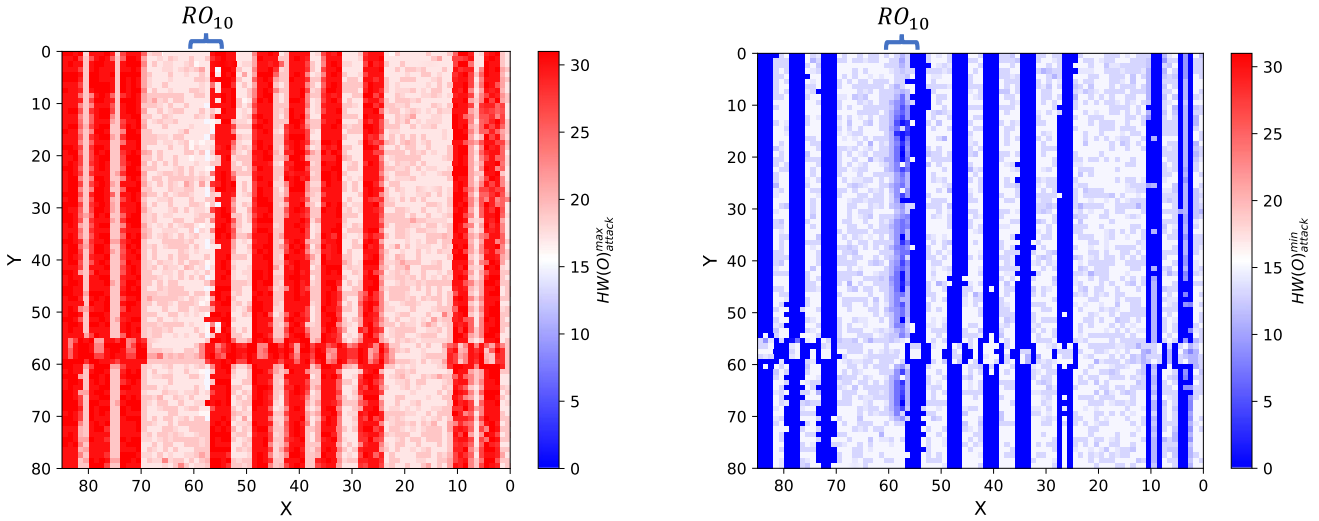
6.5 Overhead

Here, we discuss the overhead of the extended sensor for the AES.

Table 5 summarizes the logic resources consumed by the proposed AES with sensors and AES without sensors used in the experiments. When the AES and sensor were deployed, the number of LUTs increased by a factor of three and the number of FFs by a factor of 2.45. At first glance, this is the same or even higher overhead than that for the classical replication countermeasure. However, this overhead can be mitigated to some extent by optimizing the sensor and the

Table 4 Number of LFI detections by the proposed RO-based sensor in the laser-scanning experiment

	alarm	*fault*	*alarm&fault*	*alarm_rate*	*detection_rate*
Proposed sensor	3629	1383	24	73.7%	98.3%



(a) The Heat map of $Max(HW(O_{10}))$ (b) The Heat map of $Min(HW(O_{10}))$

Fig. 20 Heat map showing $HW(O)$ of RO_{10} at each irradiation

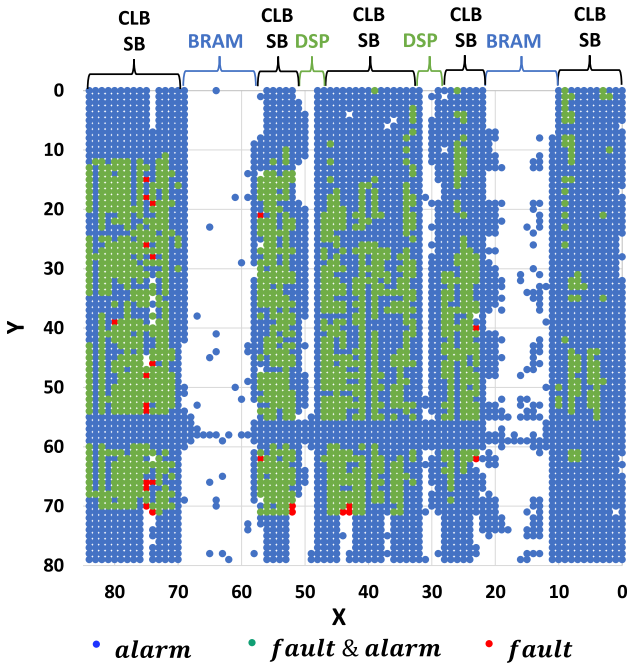


Fig. 21 Fault mapping of the AES on the proposed RO-based sensor

circuitry to be protected. When implementing the proposed sensor, we can reduce the number of AES slices that should be protected by exploiting the synthesis and placement constraints and selecting critical slices using the SoFI framework

Table 5 Comparison of consumed logic resources

AES	LUT	FF	SLICE
Unprotected	1890	611	560
On proposed sensors	4823	1499	2020

for FPGAs, similar to the case in [14]. The reduction in the number of slices to be protected not only reduces the logic resources occupied by the RO, but also reduces the logic and routing resources occupied by the delay factor because the oscillation period is shortened.

The clock speed of one round of unprotected AES is 166.8 MHz, whereas the clock speed of AES using the proposed placement constraints is 163.9 MHz. The performance degradation of the AES due to the addition of placement constraints is approximately 1.74%, indicating that the adverse impact on the critical path of the protected circuit is limited.

7 Conclusions

In this study, we clarified the factors to consider in designing digital sensors for LFI detection and introduced a methodology to design the layout of the sensors to achieve effective LFI detection. In the first experiment, we profiled the signal delay fluctuations during laser irradiation on a Kintex-7 FPGA and

clarified the relationship between the laser irradiation position and the oscillation frequency shift. Subsequently, we proposed placement constraints for the digital sensor to detect LFI into CLBs and presented both RO-based and TDC-based sensor designs based on our methodology. These two sensors were implemented on Kintex-7, and their effectiveness was verified through detailed laser-scanning experiments considering both intensity and pulse width. Furthermore, to apply the proposed RO-based sensor for AES protection, a new RO-based sensor with a carry chain integrated into the RO was proposed. The new RO-based sensor detected LFI in 73.7% of the scanning area and 98.3% of the valid faults that occurred in the AES. These results indicate that the proposed RO-based sensor with placement constraints and a carry chain is effective against LFIs in real scenarios. Our findings will provide insights for improving the tamper resistance of systems in operation or scheduled for implementation in FPGA devices for end users.

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